

Question	Answer	Marks	Guidance
1(a)	work done per unit mass	B1	Ratio must be clear. Allow 'energy' for 'work done'. Ignore symbols unless they are defined. Ignore references to units. Not any other type of named energy.
	work done (on mass) <u>moving</u> mass from infinity (to the point)	B1	Allow work done by mass moving to infinity.
1(b)(i)	potential is (defined as) zero at infinity	B1	Allow 'it' / ϕ / GPE = 0 at infinity
	work is done on (two) masses to move them apart from each other or work is done by (two) masses as they move towards each other	B1	Allow 'mass and planet' for 'two masses'. Allow ' <u>increasing</u> the separation of' for 'move them apart'. Allow ' <u>decreasing</u> the separation of' for 'they move towards each other' Allow 'potential energy of (two) masses decrease as they move toward each other' Allow 'potential energy of (two) masses increase as they move away from each other'
1(b)(ii)	$M = \phi R / G$	M1	Allow with a minus sign. Allow ϕ for Φ .
	G is the (universal) gravitational constant	A1	Ignore references to numerical value. Ignore 'gravity constant'
1(b)(iii)	$g_0 = \phi / R$	A1	Allow with a minus sign. Allow ϕ for Φ . No ECF of wrong expression in 1(b)(ii)

Question	Answer	Marks	Guidance
1(b)(iv)	curve in at least one quadrant passing through $(\pm R, \pm g_0)$ and $(\pm 2R, \pm \frac{1}{4}g_0)$ and a non-zero value of g at $\pm 3R$	B1	Ignore anything drawn in the $-R$ to R region. Allow $\pm \frac{1}{2}$ small square tolerance vertically.
	curve between $-3R$ and $-R$ of increasing magnitude of gradient and magnitude of g and curve between R and $3R$ of decreasing magnitude of gradient and magnitude of g	B1	Curves must get to within 2 small squares horizontally of $-3R$ and $3R$, and, if horizontal, horizontal for no more than 4 small squares. If in doubt, use ruler to test curvature between $\pm R$ and $\pm 3R$, $\pm R$ and $\pm 2R$, and $\pm 2R$ and $\pm 3R$. Drawn line must be entirely on the correct side of the ruler, and not touching it apart from at the end points.
	two lines drawn, one in the top left quadrant, the other in the bottom right quadrant	B1	Do not allow if lines are drawn in either of the other two quadrants.